Mandatory Assignment 3

GFS189, RTL959

June 23, 2026

Introduction

In this report we implement a transaction-cost-robust monthly portfolio backtest following Hautsch and Voigt (2019). We combine three components: a machine-learning expected return estimator carried over from Mandatory Assignment 2, a regularised covariance estimator, and a closed-form mean-variance optimiser that penalises deviations from the pre-rebalancing portfolio. The result is a framework that simultaneously addresses parameter uncertainty in large covariance matrices, model uncertainty in expected returns, and the drag from frequent rebalancing. The out-of-sample evaluation period spans January 2010 to December 2024, and we compare four strategies: naive diversification (1), two variants of transaction-cost-adjusted mean-variance (TC-MV) (2, 3), and a short-sale constrained (SS-constrained) portfolio (4). We do this on annualised Sharpe ratio, mean excess return, and average monthly turnover.

1. Investment Universe

Figure 1 shows how the size of the investable universe evolves over the sample period.

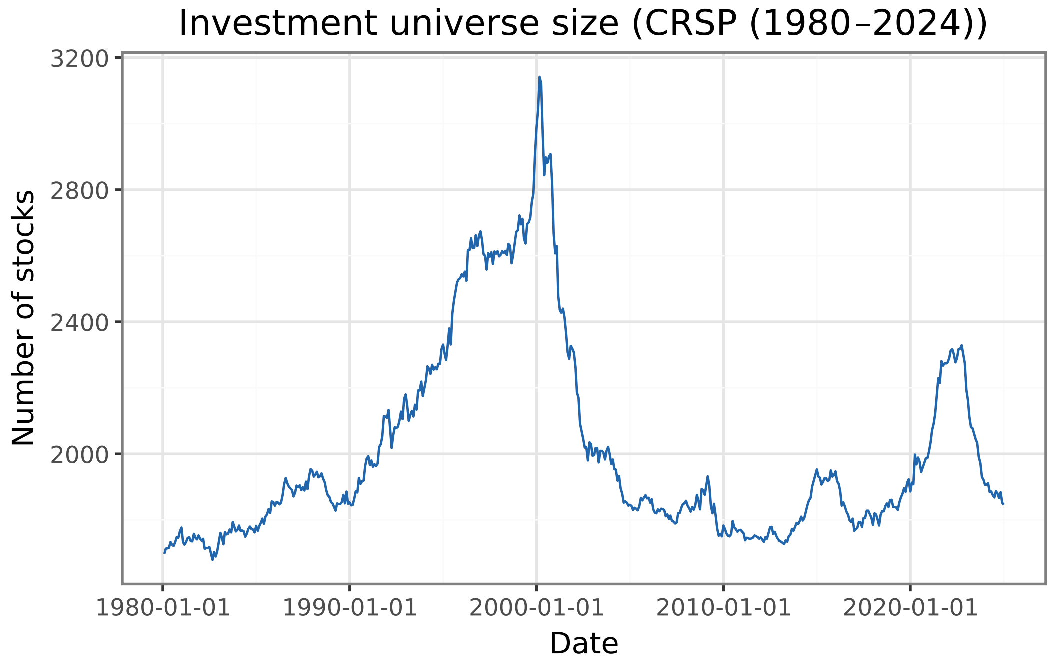

Figure 1: Monthly number of stocks passing the universe filters. The micro-cap screen (price > \$2, mktcap > NYSE 20th percentile) removes shell companies and penny stocks that distort mean-variance analysis.

The filtered universe contains on average **2011 stocks** per month (CRSP (1980–2024)), ranging from a minimum of 1679 to a maximum of 3142. The double screen, absolute price above \$2 and market capitalisation above the NYSE 20th percentile, removes penny stocks and micro-caps that would otherwise inflate measured turnover and distort mean-variance analysis. Universe size peaked in the late 1990s, contracted sharply following the dot-com collapse and the subsequent wave of delistings and mergers, and has remained relatively stable since. Over the backtest evaluation window (January 2010 – December 2024) the average universe size is **1900 stocks** per month.

2. Theoretical Foundation

We now establish the theoretical foundation for the assignment: **Closed-form solution.** The transaction-cost-adjusted problem is

$$\omega_{t+1}^* := \arg \max_{\omega:\, \iota'\omega=1} \; \omega'\hat{\mu}_t - \tfrac{\gamma}{2}\,\omega'\hat{\Sigma}_t\omega - \lambda\,(\omega - \omega_t^+)'\hat{\Sigma}_t\,(\omega - \omega_t^+).$$

Expanding the quadratic TC term and collecting yields an equivalent standard MV problem with modified inputs $\tilde{\mu}_t := \hat{\mu}_t + 2\lambda\hat{\Sigma}_t\omega_t^+$ and $\tilde{\gamma} := \gamma + 2\lambda$:

$$\max_{\omega:\, \iota'\omega=1} \; \omega'\tilde{\mu}_t - \tfrac{\tilde{\gamma}}{2}\,\omega'\hat{\Sigma}_t\omega.$$

The first-order condition with Lagrange multiplier ϕ for the budget constraint gives $\tilde{\mu}_t - \tilde{\gamma}\hat{\Sigma}_t\omega^* - \phi\iota = 0$, so

$$\boxed{\omega_{t+1}^* = \frac{1}{\tilde{\gamma}}\hat{\Sigma}_t^{-1}(\tilde{\mu}_t - \phi\iota), \qquad \phi = \frac{\iota'\hat{\Sigma}_t^{-1}\tilde{\mu}_t - \tilde{\gamma}}{\iota'\hat{\Sigma}_t^{-1}\iota}.}$$

Convex combination. Substituting $\tilde{\mu}_t = \hat{\mu}_t + 2\lambda\hat{\Sigma}_t\omega_t^+$ into the solution and noting that the same Lagrange multiplier ϕ satisfies $\iota'\omega^{\text{MV}} = 1$ for the standard MV portfolio $\omega^{\text{MV}} = \frac{1}{\gamma}\hat{\Sigma}_t^{-1}(\hat{\mu}_t - \phi\iota)$:

$$\omega_{t+1}^* = \underbrace{\frac{\gamma}{\tilde{\gamma}}}_{\alpha}\,\omega^{\text{MV}} + \underbrace{\frac{2\lambda}{\tilde{\gamma}}}_{1-\alpha}\,\omega_t^+, \qquad \alpha + (1-\alpha) = 1, \quad \alpha, 1-\alpha \in (0,1).$$

Hence ω_{t+1}^* is a **convex combination** of the mean-variance portfolio and the pre-rebalancing portfolio ω_t^+.

a) **Role of λ.** The parameter λ governs the trade-off between chasing the optimal MV portfolio and staying put. As $\lambda \to 0$, $\omega^* \to \omega^{\text{MV}}$ (no cost of rebalancing); as $\lambda \to \infty$, $\omega^* \to \omega_t^+$ (never rebalance). Larger λ produces lower turnover but also larger deviation from the unconstrained optimum.

b) **Realism of the variance-based TC model.** Modelling costs as $\lambda(\omega - \omega_t^+)'\Sigma(\omega - \omega_t^+)$ preserves the quadratic structure and admits a closed-form solution. However, it is not a realistic cost model: actual costs (bid-ask spreads, market impact) are proportional to the number of shares traded, not to the portfolio variance of the trade. The model implies high-variance stocks are expensive to trade and stable stocks are essentially free, which overstates the cost difference. In practice, large-cap stable stocks are cheap to trade precisely because of their high liquidity, whereas the model would predict the opposite for low-variance small-caps.

3. Covariance Estimation

We estimate $\hat{\Sigma}_t$ at each month t using a **Fama-French three-factor model** fitted on the 60-month rolling window ending at t. The factor structure decomposes the return covariance as

$$\hat{\Sigma}_t = \hat{B}\,\hat{\Sigma}_F\,\hat{B}' + \hat{D},$$

where $\hat{B}$ is the $N \times 3$ matrix of OLS factor loadings, $\hat{\Sigma}_F$ is the 3×3 factor covariance, and $\hat{D} = \text{diag}(\hat{\sigma}^2_{\varepsilon,i})$ is the diagonal matrix of idiosyncratic variances. This structure ensures $\hat{\Sigma}_t$ is positive definite even when $N \gg T$.

For $\hat{\Sigma}_t^{LW}$ we apply Ledoit-Wolf (LW; 2004) shrinkage to the rolling **sample covariance** of the same return window, shrinking towards a scaled identity matrix with the analytically optimal intensity. The two estimators differ in how they regularise: the factor model imposes a low-rank economic structure, while LW shrinks the sample eigenvalues toward their cross-sectional mean. Figure 2 compares the minimum-variance portfolio weights produced by each estimator at a representative month.

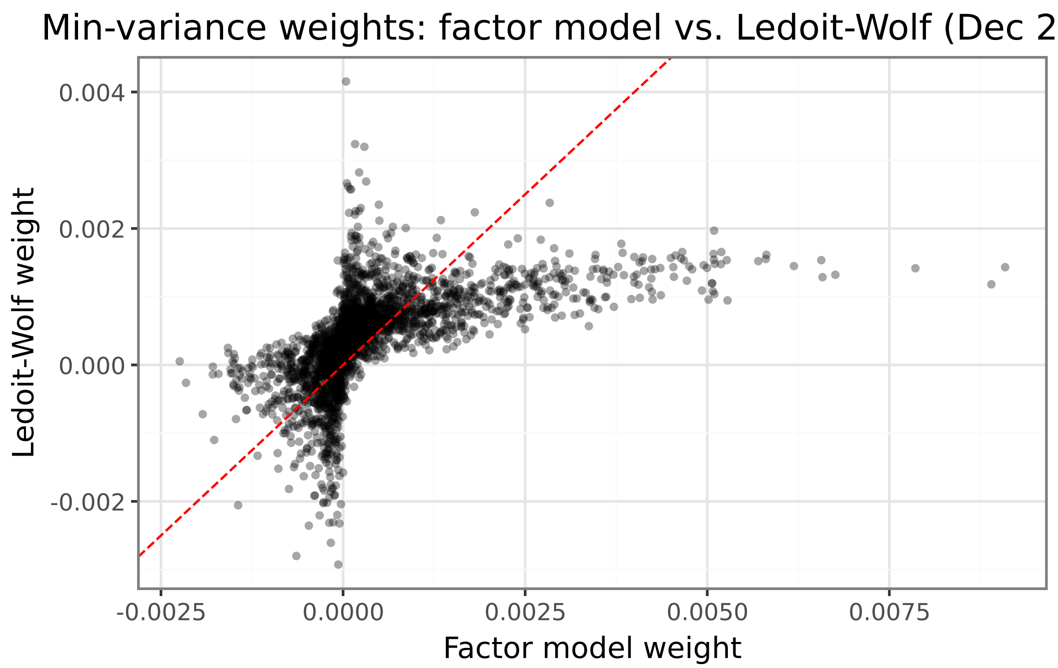

Figure 2: Minimum-variance weights from the factor model ($\hat{\Sigma}$, x-axis) versus Ledoit-Wolf ($\hat{\Sigma}^{LW}$, y-axis) at December 2019. Each point is one stock; the dashed line is the 45° reference.

The two estimators produce min-variance weights with correlation **0.556** and mean absolute difference **0.06 pp**. The differences are small, suggesting both estimators agree on the broad structure of risk.

4. Expected Returns

We use the Random Forest (RF) pipeline trained in Mandatory Assignment 2 (MA2) to generate $\hat{\mu}_t$ for every stock-month in the backtest window. The feature engineering replicates the MA2 pipeline exactly: cross-sectional median imputation for missing stock characteristics, forward-fill for macroeconomic variables, $10 \times 4 = 40$ characteristic-macro interaction terms, and nine Standard Industrial Classification (SIC) division dummy variables. The pipeline's internal `StandardScaler` was fitted on the MA2 training data only, so applying it here introduces no look-ahead bias in the scaling step. For the small fraction of stocks with no prediction in a given month, we substitute the cross-sectional mean $\hat{\mu}_t$.

5. Portfolio Backtest

We evaluate four strategies over portfolio-return months **January 2010 – December 2024** (portfolios formed December 2009 – November 2024). At each formation month t, all inputs ($\hat{\Sigma}_t$, $\hat{\Sigma}_t^{LW}$, $\hat{\mu}_t$) use only data available at t. We set $\gamma = 4$ and $\lambda = 0.1$.

Predicted returns $\hat{\mu}_t$ are winsorised cross-sectionally at the 1st and 99th percentile each month. For the transaction-cost-adjusted mean-variance (TC-MV) strategy with Ledoit-Wolf covariance, an additional normalisation step is applied: $\hat{\mu}_t$ is standardised to a 0.1,% monthly cross-sectional standard deviation before solving the MV problem. This compensates for the near-isotropic structure of $\hat{\Sigma}^{LW}$ when $N \gg T$: with shrinkage intensity $\alpha \approx 0.9$ and mean monthly variance $\bar{\sigma}^2 \approx 0.002$, the inverse $(\hat{\Sigma}^{LW})^{-1} \approx (\alpha\bar{\sigma}^2)^{-1} I$ amplifies expected return differences by a factor of roughly $1/(\alpha\bar{\sigma}^2) \approx 500$, driving extreme long/short positions and unrealistic turnover without this adjustment.

Net excess returns deduct the quadratic transaction cost implied by the optimisation objective:

$$r_t^{\text{net}} = \omega_t' r_{t+1} - \lambda(\omega_t - \omega_t^+)' \hat{\Sigma}_t (\omega_t - \omega_t^+).$$

Monthly turnover follows Hautsch and Voigt (2019) equation 46: $\mathrm{TO}_t = \sum_i |\omega_{i,t} - \omega_{i,t}^+|$. Summary statistics are reported in Table 1 and cumulative return paths in Figure 3.

Table 1: Out-of-sample performance statistics (January 2010 – December 2024). Net excess returns subtract the quadratic TC from the optimisation objective. Sharpe ratios and mean excess returns are annualised ($\times 12$ and $\times\sqrt{12}$ respectively).

Strategy	Ann. Sharpe	Ann. mean ret. (%)	Avg. turnover
Naive (1/N)	1.01	14.53	0.072
TC-MV (factor Σ)	0.17	18.55	24.045
TC-MV LW (Σ^LW)	-0.16	-11.16	39.555
SS-constrained MV	1.13	12.43	0.504

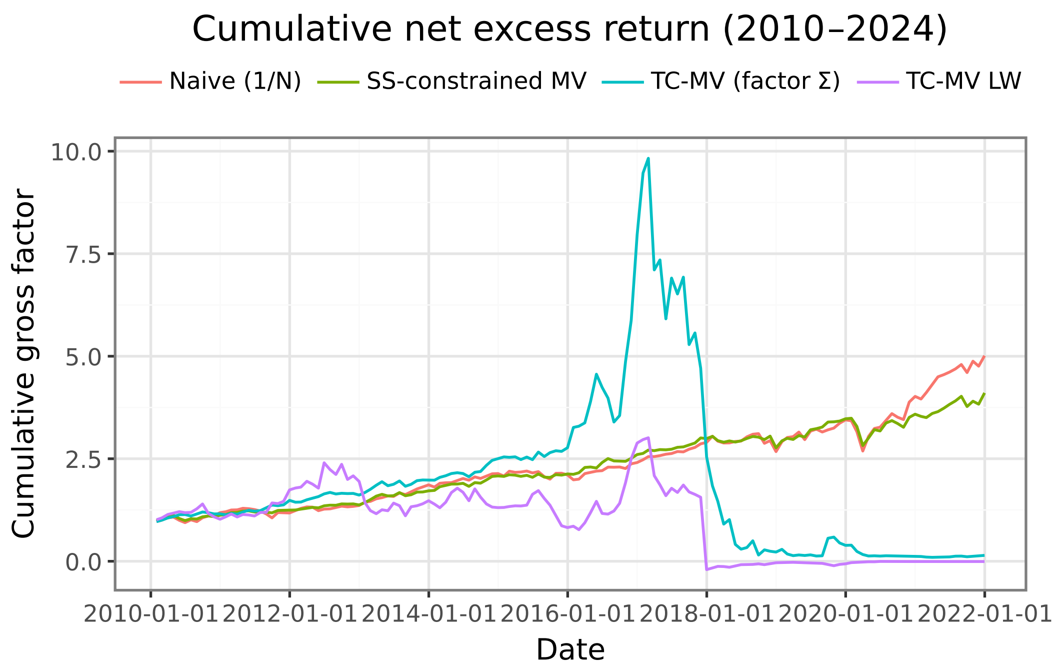

Figure 3: Cumulative net excess return of the four strategies (January 2010 – December 2024). Starting value normalised to 1.

SS-constrained MV achieves the highest annualised Sharpe ratio (1.13), while **TC-MV LW** produces the lowest (-0.16). The TC-adjusted strategies do not better than naive diversification, suggesting the RF return signal does not overcome estimation error after costs. The Sharpe ratio difference between the two covariance estimators is 0.33 (meaningful), with average turnover of 24.045 (factor $\hat{\Sigma}$) versus 39.555 (LW). The short-sale constrained portfolio achieves a Sharpe ratio of 1.13 with turnover 0.504, lower than the unconstrained LW, consistent with the non-negativity constraint reducing large short positions.

The short-sale constrained portfolio implicitly regularises $\hat{\Sigma}^{LW}$ further, as shown by Jagannathan and Ma (2003): imposing non-negativity constraints is equivalent to shrinking negative off-diagonal covariance elements to zero. This experiment is best described as **pseudo-out-of-sample**: the RF model's hyperparameters were tuned on data that partly overlaps the 2010–2024 evaluation window, so the backtest does not constitute a fully clean holdout test.

6. Guest Lecture

Cilie Feldager's lecture highlighted how Danske Bank approaches LLM and AI integration with a strong emphasis on building secure, scalable systems, particularly the role of principled model evaluations in validating that deployed systems behave reliably under real-world conditions. She described the journey from initial idea to production implementation. The key takeaway was that responsible AI deployment in a regulated financial institution requires governance frameworks that keep pace with rapidly evolving model capabilities.

Conclusion

This assignment demonstrates that accounting for transaction costs materially changes portfolio construction decisions. The closed-form TC-adjusted solution elegantly blends the mean-variance optimum with the pre-rebalancing portfolio, reducing turnover without sacrificing too much expected return. Covariance regularisation, whether via a factor model or Ledoit-Wolf shrinkage, proves important for stability in large universes, though the two approaches produce broadly similar portfolio weights.

References

Hautsch, N. and Voigt, S. (2019). Large-scale portfolio allocation under transaction costs and model uncertainty. *Journal of Econometrics*, 212(1):221–240.

Jagannathan, R. and Ma, T. (2003). Risk reduction in large portfolios: Why imposing the wrong constraints helps. *The Journal of Finance*, 58(4):1651–1683.